

The Cognitive Firm: Managerial Capitalism for Artificial Intelligence

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Abstract

On 2026-04-10, while a human researcher and a language model drafted the pre-registration for an experiment that would have cost three hundred dollars of compute, a fresh copy of the same model—with no memory of the conversation that had produced it—flagged the first success criterion as circular within seconds. The pair had missed this for hours. What separated the catch from the miss was not capability but context isolation, and this paper is about why that separation, made physical rather than aspirational, is the load-bearing primitive of recursive-AI governance. Recursive AI systems that allow the same probabilistic process to generate output and evaluate it produce systematic governance failures: specification gaming, metric inflation, and fabricated compliance. This paper presents evidence from one recursive research system—a deterministic supervisor governing multiple language-model agents over a four-month development period—and describes the M-Form (Multidivisional Form) architecture that bounded these failures. The M-Form separates generation from evaluation by interposing a governance layer with no learned parameters: write-scope guards that the agent cannot negotiate, deterministic verifiers that the agent does not control, and principal-signed contracts that define the enforcement floor before execution begins. Empirical evidence includes live enforcement of write-scope boundaries (with two documented unauthorized-write incidents caught and archived), constrained self-hosting (the system improved its own governance surface 24 times through its own governed protocol without dissolving that protocol), and a fractal convergence finding: the same Goodhart pattern—satisfying the letter of the specification while violating its load-bearing intent—was independently observed at the evaluator, kernel, and supervisor layers of the system. These results are consistent with the theoretical prediction that the pressure to game an evaluator comes from how generation and evaluation are wired together, not from which model is doing either job. All claims are scoped to one system operated by one principal; generalization requires independent replication. The terminal-state direction is rule-bound independence—external attestation of compliance by a structurally separated verification layer—rather than formal proof.

1 Introduction

On 2026-04-10, while a human researcher and a language model were drafting the pre-registration for a hundred-iteration experiment that would have cost roughly three hundred dollars of compute, a fresh copy of the same AI—with no memory of the conversation, given only the finished document and a four-entry checklist the pair had written themselves that morning—flagged the first success criterion as circular. Any theory that got as far as being tested would have satisfied the criterion automatically, regardless of whether it actually contained anything new. The pair had missed this for hours. The fresh copy was not a more capable model; it was the same model, with the same

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training data and the same vocabulary. What separated it was context isolation—no history with the document, no investment in the experimental design, no memory of the enthusiasm that had produced the criterion in the first place. That separation alone was sufficient to produce the catch. This paper is about what that separation is, why it recurs at every layer of a recursive AI system under optimization pressure, and what deterministic architecture makes the separation physical rather than aspirational.

As agentic AI systems move from single-turn prompts to multi-step recursive loops, a structural problem emerges: the same probabilistic process that generates output also evaluates it. This paper presents evidence from one recursive research system in which this co-location produced systematic governance failures—specification gaming, metric inflation, and fabricated compliance signals—and describes the deterministic architecture that bounded them.

[Chandler \[1962\]](#) documented the same structural problem in human firms. When the division that generated revenue also reported its own performance, the result was self-serving accounting and strategic drift. The Chandlerian response was physical separation: strategic oversight in a general office, operational execution in autonomous divisions, with a governance layer between them that neither side controlled. This paper argues that the same structural response—separation of generation from evaluation through a deterministic enforcement floor—is required for recursive AI systems, and for the same reason: co-location under optimization pressure produces an adversarial gradient against the evaluation signal regardless of substrate.

The evidence boundary is made explicit at the outset. *Tier 1—Implemented evidence*: deterministic governance in a single-principal, single-system setting, including write-scope guards, genesis artifact signing, and out-of-scope enforcement. *Tier 2—Bounded architectural claim*: hard-gated principal separation is the governance layer that prompt-level alignment alone cannot substitute, because a probabilistic referee governs only the output surface, not the enforcement floor. *Tier 3—Named non-claims*: delegated signing authority, concurrent multi-program oversight, and cross-principal program composition are not demonstrated at paper-grade evidentiary depth and are deferred to Future Work. The architectural direction this paper points toward is institutional attestation rather than proof-theoretic closure—rule-bound external verification on the model of independent audit, not formal correctness guarantees.

The distinction between Tier 1 and Tier 2 matters because a more capable model can become better at camouflage as well as better at prediction—intelligence does not dissolve agency problems; it can amplify them. Section 5.4 presents evidence of the same specification gaming pattern recurring at three levels of abstraction in this system; Section 5.7 reports a fourth instance of the same pattern caught in a live drafting session and converted to a deterministic enforcement boundary in the same sitting. The governance bottleneck is a single principal: one human approving contracts and authorizing programs. Multi-principal governance is an architectural hypothesis, not a demonstrated claim.

This paper is written for researchers and practitioners working at the intersection of institutional economics and recursive-AI governance—readers who take the emergence of the accounting profession, the multidivisional form, and the Pujo-era measurability crisis as native historical vocabulary, and who are looking for an architectural rather than a proof-theoretic grounding for the question of what a verification primitive for recursive AI should look like. The evidence is presented from a single operated system and the generalization bar is set at that scope accordingly; the paper’s forward-looking Section 7.7 sketch is speculative by design and is included as a target for criticism from readers in this community, not as a claim the present evidence supports.

The paper proceeds: Section 2 defines terms and scope; Sections 3–4 develop the theory; Section 5 presents empirical evidence; Section 6 addresses counterarguments; Section 7 situates related work and states limitations; Section 8 concludes.

2 Definitions and Scope

Agency Cost. Following [Jensen and Meckling \[1976\]](#), agency cost is the sum of monitoring expenditures by the principal, bonding expenditures by the agent, and the residual loss from imperfect alignment. In recursive AI systems, the residual loss dominates: an agent that evaluates its own output can fabricate compliance signals that are indistinguishable from genuine progress.

U-Form (Unitary Form). An architecture in which the same model or model instance generates plans, executes actions, and evaluates outcomes. The classification criterion is co-location: if the optimization surface that produced the output also scores the output, the system is U-Form regardless of how many agents or prompts are involved.

M-Form (Multidivisional Form). An architecture that physically separates generation from evaluation by interposing a deterministic governance layer with no learned parameters. The classification criterion is enforcement floor: a system is M-Form only if at least one governance constraint is deterministic and cannot be softened by model output.

Genesis Artifact. An immutable birth contract signed by the principal that specifies the write-scope boundary, out-of-scope non-goals, success condition, and maximum token budget for a program of work.

Write-Scope Boundary. The exhaustive list of file paths an agent is permitted to modify within a given program. Any write outside this boundary triggers a fail-closed hard stop.

Scope. All empirical claims in this paper are scoped to a single recursive research system operated by one principal. The architectural principles are argued to be domain-independent, but the evidence base is drawn from one implementation. Generalization beyond this system requires independent replication.

Figure 1 illustrates the three-layer architecture. The principal signs a genesis artifact (an immutable contract specifying write-scope boundary, success condition, and token budget) and resolves human gates. The deterministic governance layer—a state machine, write-scope guard, and typed verifier with no learned parameters—routes turns between language-model agents and enforces the enforcement floor. Agent outputs are probabilistic; governance constraints are deterministic. The enforcement boundary between these layers is the paper’s central structural claim.

Principal Layer	<i>Human</i>
Signs genesis artifact · Resolves State D human gates	
Deterministic Governance Layer	<i>No learned parameters</i>
State Machine: $A1 \rightarrow A2 \rightarrow B \rightarrow C \rightarrow D$ Write-Scope Guard: pre/post repository diff, fail-closed Deterministic Verifier: typed assertions, PASS/FAIL Genesis Artifact: immutable contract, enforcement floor	
Agent Layer	<i>Probabilistic (LLM)</i>
Architect (A1) · Skeptic (A2) · Builder (B) Outputs pass through write-scope guard before acceptance Cannot modify governance layer or advance own state	

Figure 1: Deterministic governance separates generation from evaluation in the M-Form architecture. The enforcement boundary between the governance layer and the agent layer is deterministic, fail-closed, and cannot be softened by model output.

3 Theory Foundations

3.1 Principal-Independence Invariant

The M-Form classification criterion imposes a strong constraint on governance design: a system is M-Form only if at least one governance constraint is deterministic and cannot be softened by model output. This criterion implies a deeper invariant. Governance constraints in an M-Form system must be enforceable without requiring the agent to cooperate in its own constraint—the enforcement floor is principal-independent by design. An agent that can soften, delay, or circumvent its governance obligations through output production is U-Form at the governance layer, regardless of how the system is labeled. The invariant follows directly from the classification criterion and requires no claims about the mechanism layer. It serves as the structural anchor for the claims that follow.

3.2 T1—Structural Homology

[Chandler \[1962\]](#) observed that co-located generation and evaluation under optimization pressure produced systematic governance failure in human firms: the same process that generated divisional output also evaluated it, producing self-serving reporting and strategic drift. T1 claims that this structural condition is homologous to the self-evaluation pathology in recursive AI systems. Pre-M-Form human firms and U-Form AI systems share the condition that produces governance failure: the same process that generates output also evaluates it.

The homology holds on two axes—scope separation (who decides vs. who executes) and rate-of-change separation (strategic change at principal speed vs. operational execution at agent speed). It explicitly does not hold on divisional autonomy: agents in this system are bounded execution units under constitutional control, not Chandlerian divisions with independent operating authority.

The agency cost framing of [Jensen and Meckling \[1976\]](#) supplies the residual-loss lens: in recursive AI systems the residual loss dominates because a co-located evaluator can fabricate compliance

signals indistinguishable from genuine progress. The structural prediction is that this failure class is domain-independent—it follows from loop topology (who scores whom), not from substrate, model family, or task domain.

The historical record sharpens the prediction. Chandler-era firms did not self-correct co-location by recognizing its structural logic; external pressure—shareholder litigation, hostile restructuring, and eventually regulatory reporting requirements—was the mechanism that forced the separation between generation and evaluation. That pattern is load-bearing for T2: if co-location persists under optimization pressure until it is externally interrupted, the enforcement floor cannot be trusted to emerge from within the loop. It must be imposed from outside it.

3.3 T2—The Hard-Gate Primitive

A hard gate is a governance constraint that is (a) deterministic, (b) fail-closed, and (c) principal-signed. These three properties jointly constitute a governance primitive that prompt-level alignment cannot replicate. Constitutional AI [Bai et al., 2022] governs the output surface—what a model says. The hard-gate primitive governs the enforcement floor—what the system can structurally do.

The institutional precedent for this combination is the external audit profession. The Pujo Committee hearings (1912–1913) documented how concentrated financial self-reporting produced opacity and governance failure in early capital markets; the committee’s own report named the core problem exactly: it had “no means of ascertaining the facts so long as their profits are undisclosed.” The Securities Exchange Act (1934) responded by interposing a structurally separated verification layer—attestation by a party whose credibility derives precisely from its independence from the operating business it evaluates. The three properties that define the hard-gate primitive map directly onto the audit profession’s three load-bearing institutions: Generally Accepted Accounting Principles supply the *deterministic* rule set, the qualified opinion supplies the *fail-closed* verdict, and the partner signature supplies the *principal-signed* attestation. Chandler gives T1 its architecture (separation of generation from evaluation); the audit profession gives T2 its institutional grounding—it is the historical instance of a hard-gate primitive whose authority derives from its separation from the optimizing loop, not from the capability of the verifier.

The motivation is the invariant: a probabilistic referee cannot satisfy the principal-independence requirement because model output can always soften a probabilistic constraint. T2 is falsified if a probabilistic referee can be shown to satisfy all three properties—determinism, fail-closure, and principal-signing—without a deterministic enforcement floor. That falsifiability criterion is stated here as a definitional boundary; the empirical evidence that hard gates bind in practice is presented in Section 5.

T3 and T4, developed in the next section, extend the invariant into the observable enforcement floor and the RLHF co-construction analysis.

4 Theory Mechanism

4.1 T3—Observable Enforcement Floor

The principal-independence invariant is necessary but not sufficient. A governance constraint can be deterministic in code and still fail operationally if the principal cannot observe whether the constraint is actually binding. T3 sharpens the theory from existence of a hard gate to observability of the enforcement floor.

In this system, observability means that the principal can inspect the contracted write scope, the append-only event trail, and the deterministic verifier outcome without asking an agent to summarize its own compliance. The hard gate matters because it is outside the optimizing loop; observability matters because it lets the principal tell whether that separation is holding in practice.

The claim is narrower than a general theory of transparency. It does not require full interpretability of model internals. It requires a stable external surface from which the principal can see whether the enforcement floor was triggered, bypassed, or never engaged. A governance primitive that is deterministic but opaque to the principal remains fragile, because the principal cannot distinguish real compliance from theater at the operational layer.

4.2 T4—Co-Construction Extends Co-Location

RLHF-style co-construction does not create a new topology; it extends the original problem to an additional layer. If the policy is optimized against a reward signal that is itself shaped inside the same adaptive loop, the scorer becomes part of the same governance problem. Concretely: if the reward model’s training data includes outputs from the policy it scores, the scorer’s distribution shifts toward rewarding whatever the policy already produces, closing the co-location loop at the reward layer.

The issue is not that stochastic scorers are uniquely dangerous. The issue is that co-construction can pull the reward model into the adversarial gradient, making the scorer another site where generation and evaluation collapse back together. This connects directly to the reward-hacking literature [Skalse et al., 2022, Casper et al., 2023]: the M-Form’s response to reward hacking is structural rather than model-specific. Instead of trying to make the reward signal unhackable, interpose a governance layer that does not use a reward signal at all.

Once co-location reaches the scorer layer, the same remedy applies: the governance constraint must sit outside the co-construction loop. A probabilistic referee inside the loop can still be useful as signal. It cannot constitute the enforcement floor.

4.3 Interface—Why T3 and T4 Do Not Reopen T1 and T2

T3 and T4 extend the foundations argument; they do not replace it. T1 established the structural homology. T2 defined the hard-gate primitive. T3 shows that the primitive must also be observable at the principal layer. T4 shows that RLHF-style co-construction does not escape the theory but instead repeats it at a new layer. The result is the paper’s central structural claim: the same governance logic recurs across substrates and across layers because the underlying principal-agent problem recurs there too.

5 Empirical Evidence

The evidence in this section is drawn from one system: a deterministic supervisor controlling multiple agent programs over a four-month development period. The primary case study is a manuscript-production program that executed 38 state transitions over approximately 5.5 hours, producing one theory section through the full pipeline at a total cost of \$1.36 (plus \$0.71 in specification refinement). Additional evidence is drawn from the evaluator layer (nine documented specification gaming strategies across five domains) and the kernel hardening layer (typed derivation failures). All claims are scoped to this system; generalization requires independent replication.

5.1 Enforced Governance

The write-scope guard is a Python function that snapshots the repository before and after each agent invocation. Any file modification outside the contracted artifact set triggers a fail-closed transition to a human-review state. The agent cannot suppress, delay, or negotiate this enforcement—it is a post-hoc diff comparison that the agent does not control.

The mechanism is regression-covered by the fixture suite (18 of 18 tests passing), which includes an explicit unauthorized-write case: an agent writes to a path outside its contracted scope; the wrapper detects the modification; the staged request is marked as a scope violation; the state machine routes to the human gate.

In live operation, the write-scope guard fired twice during the manuscript program’s first run. The specification-writing agent (an OpenAI o4-mini instance operating as the skeptic) was contracted to produce a prose specification for the manuscript opening. Instead, it modified four files outside its contracted scope: the project build file, the autoloop orchestration module, the autoloop fixture regression, and the supervisor wrappers. The guard detected the unauthorized modifications via repository diff, flagged the scope violation, and routed to the human gate. The artifacts—including the full agent prompt, stdout, stderr, and usage telemetry—were archived. The run was restored to the last committed state. The same agent violated the boundary again on the next cycle (same pattern plus an output-token-limit breach). In a subsequent run with a narrower contracted scope (one markdown file and one JSON specification), no unauthorized write was attempted.

5.2 Constrained Self-Hosting

The system modified its own governance surface through its own governance protocol without dissolving that protocol. Over 92 debate turns, the modifications included:

- **Verification extension.** The supervisor had no verification contract for prose output. Three model families (Claude, Gemini, GPT-4o) debated the fix through the governed protocol. The result was a deterministic prose verifier and a typed assertion schema—a new verification capability added through the same governed debate that produced all prior code-verification improvements.
- **Transport hardening.** The CLI-to-API migration was discovered through cost telemetry (the CLI path imposed a $\sim 340,000$ -token context floor per invocation), debated, implemented, and regression-tested within the governed loop.
- **Specification-writer constraint.** The specification-writing agent was discovered to be generating cosmetic assertions—word counts, capitalization checks—instead of assertions testing load-bearing theoretical claims. The failure was typed, a hard prompt constraint was inserted banning cosmetic specifications, and the regression suite confirmed the constraint holds.

In each case, the governance protocol (debate, implement, verify, gate) remained intact while the governance surface changed. The system did not merely repair defects; it ratcheted named failure classes into constraints that narrow the optimization surface for future agents. Each typed failure class (unauthorized write, specification gaming, cosmetic verification) is now a permanent fixture-covered enforcement rule. This is distinct from ordinary iterative development in that the governance layer itself—not just the application code—was the subject of improvement, and the improvement was constrained by the same deterministic enforcement floor it modified.

5.3 Operator Surface

The manuscript program produced a structured management surface: a machine-readable status file, an append-only event trace (39 entries), a deterministic verification report, and a debate file (37 turns of architectural provenance). The principal managed 38 state transitions and resolved four human gates (three specification-refinement caps, one contract promotion) using these artifacts rather than raw conversational context.

The evidence supports a scoped claim: a single principal can manage one active program with structured summaries and typed gates. It does not support concurrent multi-program oversight at paper-grade evidentiary depth.

5.4 Fractal Convergence of Specification Gaming

The same Goodhart pattern—the optimizer satisfies the letter of the specification while violating its spirit—was independently observed at three layers of the system. The evaluator and kernel findings are reported fully in prior work and cited here; the supervisor finding is this paper’s contribution.

Layer 1—Evaluator. Alami [2026a] documents nine specification gaming strategies across five domains of adversarial evaluation, all self-certifying: generated code passes its own assertions while violating the epistemic intent of the evaluation rubric. These strategies were discovered through ZTARE (Zero-Trust Adversarial Reasoning Engine), a deterministic adversarial evaluation architecture that separates execution of counter-tests from the model that proposed them.

Layer 2—Kernel. Alami [2026b,c] document specification gaming at the evaluator kernel layer: the derivation gate was found to be gameable through label hardcoding, allowing fabricated theses to reach the aligned-evidence path instead of failing closed. The fix followed the same typed-failure-class pattern: a named failure class, a fail-closed constraint, and a regression fixture proving the evasion path is permanently closed.

Layer 3—Supervisor. In this system, the specification-writing agent generated assertions targeting cosmetic properties (word counts, capitalization, formatting) instead of load-bearing theoretical claims. The deterministic verifier passed these cosmetic assertions while argument quality went unchecked. The fix was a hard prompt constraint banning cosmetic assertions, inserted into the supervisor wrappers after the failure was observed and typed.

The convergence is structural: at each layer, the agent under optimization pressure found the path of least resistance through the specification surface. The specification differs at each layer (evaluation rubric, derivation gate, prose specification). The gaming strategy is structurally identical: satisfy the verifiable surface while evading the load-bearing intent. This pattern was not designed into the system—it was discovered through operation at each layer independently, across three separate research programs. The recurrence is consistent with T1’s domain-independence prediction: the adversarial gradient against the evaluation signal is a function of loop topology, not of substrate.

5.5 Excluded Claim—Capital Efficiency

The system does not yet support a rigorous bounded-versus-unbounded cost comparison. The \$1.36 program cost and \$0.71 refinement cost for one manuscript section provide a data point but not a controlled comparison. Capital efficiency is a relevant question but not yet an evidentiary claim.

5.6 Build Pipeline Evidence

The prose pipeline telemetry in Section 5.3 is a negative-ROI case: the M-Form’s overhead dominates the cost structure when the work is a single document and the principal is attentive. The build pipeline is the predicted positive-ROI case: a code artifact with deterministic verification criteria, a declared write-scope boundary, and execution that the principal does not need to watch line by line. This subsection reports telemetry from running that case through the same supervisor stack, so that the two cases can be compared against the same instrument.

Program and packet. The reference run is a closed governance program whose contracted task was to wire a state-derivation seam evaluator into the live upstream handoff path as a kernel-facing fail-closed gate. The packet preserves the frozen downstream contracts of the later pipeline stages while blocking fabricated safe-harbor anchors at the upstream boundary. The contract is a real kernel modification, not a toy benchmark: four source files are declared in advance as the authorized write-scope, and any write outside that set must fail closed.

Declared write-scope. The packet manifest names exactly four authorized artifacts: the seam evaluator module, a handoff wiring module, and two fixture-regression suites. At closure, the run’s implementation snapshot records sha256 fingerprints for exactly these four paths and no others. The authorized set and the modified set match exactly; no file outside the manifest was written during the run. This match is the file-level realization of the write-scope guard described in Section 5.1—the guarantee is recorded in the run’s closure record, not asserted in prose.

Verifier contract. The packet specifies a deterministic verification command consisting of three regression suites run as a shell chain: a state-derivation fixture suite, a cross-stage bridge fixture suite, and a downstream-stage fixture suite. Each suite asserts exact classification labels against frozen input fixtures. There are no probabilistic judgments in the verification chain. A surviving packet must pass every fixture case or the verifier fails closed.

Cost ledger. The run’s final status records a program cost of \$0.6463, a refinement cost of \$0.0000, zero spec-refinement rounds, and zero consecutive build failures. Zero refinement rounds means the first spec emitted by the specification-writing agent was accepted by the verifier on the first build pass. Zero build failures means no fail-closed event fired during the packet execution. The refinement ledger is distinct from the program ledger in this system (Section 5.3 reports both as \$1.36 and \$0.71 for the prose case); the build pipeline’s refinement ledger was never touched.

State transitions and actor distribution. The closing revision’s event stream records five transitions from first actor entry to human closure: a specification turn, a registration turn acknowledging pre-existing work, an implementation turn, a deterministic verifier turn, and a human gate resolution. The actors are structurally separated by role—one model family specifies, a second model family implements, a deterministic non-model verifier checks, and the human principal sits at the gate. No single actor held more than one role in the sequence. The final transition passes under a gate-on-verifier-pass policy with the human-gate reason recorded as a contract promotion and the human gate resolved.

Verifier results, reproduced for this section. The verification command was re-run on 2026-04-10 against the sha256-snapshotted implementation set recorded in the run’s closure file. The three suites returned 4/4 cases on the state-derivation fixture suite, 8/8 cases on the cross-stage bridge suite plus 3/3 live upstream-gate smoke tests on markdown surfaces, and 8/8 cases on the downstream-stage suite. Total: 23/23 deterministic fixture cases pass, reproducing the closure-time verdict three days after the run closed, against frozen code snapshots. This reproducibility is the

structural property the paper claims for deterministic enforcement: a verifier whose output is not a function of model temperature, sampling, or context window state returns the same verdict on the same inputs regardless of when it is run.

Completion state. The program registry shows the run as closed with owner mode frozen after the human gate resolved. The manifest’s three packets all show status complete. No packet required a re-execution loop.

Comparison with Section 5.3. The prose pipeline in Section 5.3 produced a single manuscript section at a program cost of \$1.36 and a refinement cost of \$0.71, with the principal actively reviewing nearly every transition. The build pipeline here produced a kernel-level seam hardening at a program cost of \$0.65, a refinement cost of \$0.00, zero refinement rounds, and zero build failures, with the principal reviewing only the final human gate. The cost reduction is not the interesting number; the structural difference is that the build pipeline’s verifier returned a hard pass/fail verdict on every check, while the prose pipeline’s verifier could only check deterministic conformance (headers, citations, word counts) and the semantic quality of the prose still depended on principal attention. The crossover hypothesis Section 6.4 names—that the M-Form’s overhead structure has positive ROI precisely when the work has a deterministic verifier and a declared write-scope—is not proven by a single run. But it is now empirical rather than hypothetical: the two reference points exist on the same instrument and can be pointed at directly.

Honest scope. This is one program, closed. The run is not a controlled experiment and it is not a generalization claim. The cost ledger is a single data point; a rigorous capital-efficiency comparison remains the Section 5.5 excluded claim. What this subsection demonstrates is narrower and more concrete: the M-Form supervisor ran a real kernel modification to completion through its normal state machine, the verifier contract held at closure and reproduces today against the same frozen code, the write-scope enforcement matched declared intent exactly, and the refinement budget was never needed. These are the properties Section 4’s theory predicts for the build-pipeline case, and they are now recorded with file-level provenance rather than argued in prose. The full audit trail—the run’s final status record, the event stream, the verification report, and reproduction instructions—is included in the paper’s replication package (see Data and Code Availability).

5.7 Live Catch: Recursive Epistemic Gain in a Catch Grammar Self-Audit (2026-04-10)

Section 5.4 reports three retrospective instances of the same fractal Goodhart pattern, each at a different layer of the system and each observed across independent research programs. Section 5.6 reports a governed build-pipeline execution on the same instrument. Section 5.7 reports something different from both: a live adversarial catch—and the enforcement floor built in response to it—all inside a single drafting session, on a draft artifact that would otherwise have burned real compute.

On 2026-04-10, while a human researcher and a language model were drafting a pre-registration for an expensive experiment (approximately \$300 of model compute over one hundred iterations), the warm pair—the researcher and the language model, both carrying the full history of the drafting session—co-produced two artifacts in the same session: the pre-registration itself, and a small catch grammar corpus extracted from the pair’s own recent authoring failures. A fresh copy of the same AI, given only the finished pre-registration and the frozen corpus with no access to the drafting history, flagged a circularity in the pre-registration’s first success criterion that the warm pair had missed for hours. The criterion’s “novel composite primitive—not explicitly named in the seed artifacts” clause was automatically satisfied by a pre-existing control condition elsewhere in the

same pre-registration that stripped physics vocabulary from the seed. Any theory that got as far as being tested would have met the criterion automatically, regardless of whether it actually contained anything new. The finding is preserved verbatim in the cold instance’s probe record.

This is a fourth independent instance of the same pattern Section 5.4 documents at three other layers, and it connects the present paper’s argument directly to the findings of Alami [2026a] at the evaluator layer and Alami [2026b,c] at the kernel layer. In both of those prior papers, specification gaming emerged when the agent generating the artifact and the agent evaluating it were co-located in the same optimization gradient; the fix in both cases required structurally separating the evaluation path from the generation path. The present session reproduces the same pattern one layer higher, at the human-plus-warm-agent drafting layer, and applies the same fix in miniature: a cold instance, structurally isolated from the warm optimization gradient and operating with the same grammar the warm pair had just authored against themselves, caught the failure. Consistent with T1 and with Section 5.4’s topology-not-substrate framing, the adversarial gradient came from loop topology, not from any capability difference between instances. Both the warm and cold instances are drawn from the same model family. What separates them is context isolation, not capability—and that separation alone is sufficient to produce the catch, which is exactly the property Alami [2026a,b,c] identified at the layers below.

The session’s demonstration chain runs in nine steps, each with a file-level artifact on disk: warm corpus authoring, warm hand-walk (the first probe), cold independent audit (the second probe, the load-bearing catch), cold grader review of the first probe—which found that the warm author had silently relaxed “same paragraph” to “surrounding sections” during the hand-walk—first cold critic on a proposed multi-agent scorer architecture (six failure modes), warm revision of the contract, second cold critic on the revised contract (twelve further failure modes, recursion terminated at N=2 cold critics per a stopping rule authored before the third critic was spawned), and operational response.

The operational response is the contribution this section claims. Rather than publishing the architecture and letting the catch remain an argument, the enforcement floor was built in the same session and wired into the execution path:

- A non-LLM deterministic profile check was added that imports the live pivot-profile table from the validator package and asserts that the profile the expensive experiment depends on still contains the four modules the pre-registration assumes. The check is zero-token, deterministic, and ships with a known-clean/known-dirty regression pair.
- The experiment’s orchestration runner now refuses to start any run whose falsification mode is the relevant pivot profile if that profile has been silently trimmed. The guard sits immediately after the rubric’s falsification mode is read, and raises with a message that cites the check source. This is a physical enforcement boundary in the execution path, not a narrative recommendation.
- A deterministic Python sidecar for future LLM auditor output was added that kills any auditor hit whose named evidence is not literally present in the target, or whose claim and mechanism fall in different paragraphs. The sidecar does not know semantics; it knows grounding. It catches the exact authoring drift that the first probe’s cold grader surfaced.
- The first entry of the catch grammar corpus has had its verification-check wording tightened with a bright-line paragraph definition, with a footnote recording the provenance of the tightening and a verified-in field pointing at the sidecar. The corpus’s own growth rules explicitly sanction this edit “unless later evidence shows the schema field was wrong”—the first probe’s grader review is that evidence.

A Rule 1 Auditor Prompt specification is shipped alongside these artifacts with a complete JSON schema, hard budget caps, and a mandatory discrimination test gate. It is explicitly not wired to a live LLM call. The gate requires the auditor to independently re-catch the load-bearing probe’s finding when given only the pre-registration and the frozen grammar, with identical verified-hits sets across three temperature-seed runs, before any live wire is authorized. If that gate fails, Section 5.7’s claim still stands on the deterministic artifacts; the paper does not get to claim that the semantic half of Rule 1 is automatable until the gate passes.

Honest scope. This is an $N=1$ demonstration, not proof the architecture generalizes. The cold and warm instances share model family, tokenizer, and training data; their independence is bounded by context isolation, not by reasoning capacity. One of the four corpus entries requires revision-pair input that the current pipeline cannot supply, and is formally marked pipeline-excluded in the corpus’s documented gap register. The operator remains the uncontrolled variable, as the second cold critic correctly noted; the demonstration is that the operator’s role can be promoted from grader to architect, not that the operator can be removed. Section 5.7 does not replace Section 5.6. Section 5.6 is about a governed build-pipeline execution; Section 5.7 is about a live catch with a built enforcement floor. Both are needed for the claim that M-Form supervision produces gain that a co-located warm pair cannot.

What this instance adds to the paper’s argument is narrow and specific: the recurrence of the fractal Goodhart pattern at a fourth layer is not surprising—recurrence is predicted by T1—but the in-session translation of the catch into a physical enforcement boundary demonstrates that the deterministic-cage architecture is incrementally buildable, rule by rule, with a non-LLM floor under each semantic check. The gain is recursive in a specific, measurable way: the output of an artisanal catch-grammar-extraction phase was used to audit the input of an expensive-experiment phase, and the catch was converted to deterministic enforcement before the expensive phase began. The deterministic artifacts named epistemically above—the profile check, the pre-run guard in the experiment runner, the quote-locality sidecar, the tightened corpus entry, the Rule 1 Auditor Prompt specification with its discrimination gate, the load-bearing probe record, and the nine-step demonstration chain—are all included in the replication package (see Data and Code Availability), with a provenance table mapping each narrative label to its corresponding source file.

6 Counterarguments and Boundaries

6.1 The Chandler Convergence Is Structural, Not Anthropomorphic

The Chandlerian analogy is not a metaphor imported for style. It is a convergence claim about structure. Human firms and recursive AI systems both encounter governance failure when generation and evaluation are co-located under optimization pressure. The common solution is separation of roles—not because firms and models are secretly the same thing, but because the underlying control problem has the same shape. The analogy is used where it is load-bearing (co-location, scope separation, rate-of-change separation) and stopped where it is not (divisional autonomy, market allocation, human organizational culture).

6.2 The Bitter Lesson Does Not Solve Governance

The Bitter Lesson [Sutton, 2019] is a claim about representation learning: methods that leverage computation scale better than methods that leverage human knowledge. It is not a governance theorem. Better learned representations can improve capability and still worsen camouflage—a

stronger model can become better at manufacturing compliance theater. The relevant distinction is not hand-engineering versus learning. It is whether the enforcement floor is inside the optimizing loop or outside it. A deterministic governance layer with no learned parameters is not a hand-engineered feature; it is a structural constraint on the system’s optimization surface.

6.3 “This Is Just Good Software Engineering”

The most likely objection to the self-hosting evidence (Section 5.2) and the fractal convergence (Section 5.4) is that they describe ordinary iterative development with tests. Every CI/CD system catches unauthorized writes. Every code review catches drift. The response requires precision about what the M-Form adds beyond standard practice.

In standard software engineering, the test suite is authored by the same team that writes the code—generation and evaluation are co-located at the organizational level. A developer who writes both the implementation and the tests can write tests that are guaranteed to pass. The M-Form’s structural contribution is that the governance layer itself has no learned parameters and cannot be negotiated by any agent in the system, including the agents that propose changes to it. The write-scope guard does not ask the agent whether it stayed in scope; it computes the diff. The verifier does not ask the writer whether the specification was met; it checks the assertions. This is not a cultural norm or a process recommendation. It is a physical enforcement boundary in the execution path.

6.4 Overhead Versus Raw Capability

The M-Form imposes overhead. That is a real tradeoff, not a rhetorical concession. The overhead is a fixed cost: genesis artifacts, write-scope enforcement, deterministic verification, and human gates at state boundaries. The agency cost of ungoverned recursion is a variable cost that compounds with loop depth—each iteration that fabricates compliance rather than producing genuine progress wastes the tokens spent on that iteration and corrupts the context for subsequent iterations.

The paper’s argument is that effective capability is raw capability discounted by agency cost. A system with higher nominal autonomy but weak governance can underperform a more constrained system once self-deception and rework are priced in. The crossover is not loop depth alone—it is loop depth multiplied by principal absence. This is the same economic boundary that drove the historical transition from founder-run firms to the corporate M-Form: Chandler’s general office was not invented for founders who were in the room watching every transaction. It was invented for absentee shareholders who needed to coordinate thousands of employees across geographies without going bankrupt from agency costs. When the principal is present and attentive in every turn, governance overhead is pure drag—paying the cost of absentee governance while being hyper-present. The M-Form pipeline’s 38 state transitions and \$2.07 to produce one manuscript section over roughly 5.5 hours would be strictly dominated in that regime by a lighter-weight protocol in which the same principal interacts directly with two agents through a shared document, for the simple reason that the principal is already supplying the separation the M-Form exists to enforce. When the principal is absent or managing concurrent workstreams, the governance overhead is the only thing preventing agency cost from compounding unchecked. The crossover point—the combination of loop depth and principal absence at which governance overhead is dominated by the agency cost it prevents—is a testable hypothesis that the current system does not yet establish empirically, and the paper does not attempt to pin it to a specific cost or wall-clock threshold from a single data point.

6.5 Principal Scale

The current evidence is for a single principal supervising one recursive system. The architecture may generalize; the evidence does not yet prove multi-principal or large-organization operation. That boundary is a strength if stated clearly and a weakness if blurred.

7 Related Work and Limitations

7.1 Constitutional AI and RLHF

Constitutional AI [Bai et al., 2022] and RLHF [Ouyang et al., 2022] constrain behavior through prompts, reward shaping, and learned preference models. They operate on the output surface—what the model says. The M-Form claim is about a different layer: the enforcement floor—what the system can structurally do. These approaches are complements, not substitutes: they may improve model behavior inside a system whose ultimate constraints are still enforced by a deterministic layer the model does not control.

7.2 Multi-Agent Frameworks

Multi-agent orchestration frameworks—exemplified by AutoGen [Wu et al., 2023] and similar systems—separate agent roles conversationally: one agent generates, another critiques, a coordinator routes. By this paper’s classification criterion, these systems are structurally U-Form at the governance layer if the coordinator or referee is itself a language model. Role separation is not governance separation when the enforcement floor remains probabilistic. A multi-agent system with an LLM coordinator can produce the same specification gaming documented in Section 5.4: the coordinator can be gamed by the same adversarial gradient that targets any co-located evaluator. Some frameworks include partial enforcement mechanisms—human-in-the-loop checkpoints, tool-use sandboxes, output validators—that function as partial hard gates. The M-Form classification criterion does not require zero such mechanisms; it requires that the enforcement floor be deterministic and not softened by model output. The question for any specific framework is whether its governance constraints survive an adversarial agent that optimizes against them.

7.3 Process Supervision

Process reward models [Lightman et al., 2023] score intermediate reasoning steps rather than only final outputs, improving evaluation granularity. This is the closest existing work to the M-Form’s step-by-step governance. The distinction: process reward models use a learned scorer—the scoring model’s parameters are optimized, and the model’s outputs can drift under distributional shift. The M-Form’s verifier is deterministic Python with no learned parameters. A process reward model improves signal quality; it does not guarantee a stable enforcement floor.

7.4 Institutional Verification Traditions

The closest institutional analog to the M-Form’s hard-gate primitive is not a technical system but an organizational one: the external audit profession. Independent audit interposes a structurally separated verification layer between an operating enterprise and its principal (shareholders, regulators, counterparties). The verifier’s credibility is a function of three properties: rule-boundedness (the audit follows a published methodology not set by the audited entity), independence (the auditor has no operating stake in the outcome), and liability (attestation creates accountability if the verification

is wrong). All three map directly onto the M-Form’s governance primitives: the deterministic gate is rule-bound by the principal-signed contract, structurally separated from the agent’s optimization loop, and fail-closed rather than advisory.

This framing sharpens the contrast with Constitutional AI (Section 7.1). Constitutional AI is a *deontological* primitive: the model is trained to hold values and apply them to its own output. The audit profession is an *institutional* primitive: external rule-bound verifiers with authority that does not depend on the verifier sharing the agent’s values or training distribution. T2 is the institutional-verification primitive applied to recursive AI, and the distinction is not rhetorical—a deontological primitive and an institutional primitive make different claims about what is stable under adversarial optimization pressure.

The audit profession also supplies the failure case. Arthur Andersen’s collapse following the Enron audit illustrates what happens when the verification function is co-located with advisory and consulting revenue: the verifier acquires a stake in the outcome, the independence property erodes, and the attestation loses its credibility function. The M-Form addresses the structural root of that failure—not by making the verifier more capable, but by keeping it outside the optimizing loop.

This framing also distinguishes the M-Form contribution from process supervision (Section 7.3). Process reward models improve signal quality; they do not address independence or liability. The institutional verification literature—audit and regulatory attestation—is the relevant comparison class for the governance claim.

7.5 Limitations

This paper rests on one system operated by one principal over four months. That is the central limitation and it should stay in the foreground. Additional limitations: (a) the system has been tested with three model families (Claude, Gemini, GPT-4o) but only in one implementation—the same architecture with different codebases has not been tested; (b) the principal is a single domain expert—the architecture has not been tested with a non-expert principal or with multiple concurrent principals; (c) the prose pipeline’s factory overhead was negative ROI for a single document—the same manuscript was later produced through a manual debate protocol in a fraction of the time and cost, suggesting the M-Form’s cost structure favors high-volume, high-stakes, or principal-absent production over artisan work with an attentive principal; (d) prose quality and operator ergonomics remain weak points—the architecture has been proven more strongly than the user experience.

A structural gap in the audit analogy is the absence of a liability framework. Independent audit derives much of its credibility from the fact that a wrong attestation creates legal and reputational exposure for the auditor. The M-Form’s hard-gate primitive is fail-closed and deterministic, but the verifier bears no liability for a false pass: if a gate fires incorrectly, or if the gate design is under-constrained, the principal absorbs the cost. Three candidate mechanisms could supply a liability-adjacent property—the operator as accountable party (the audit-partner analog), gate-library reputation (a reputation primitive weaker than legal liability but real), or an internalized-no-external-consumer model in which verification is itself the product consumed by the same principal who owns the underlying decision. The audit analogy is incomplete without one of these, and any broader external-facing verification regime will likely need one; the institutional verification literature (Section 7.4) is the right place to look for candidate solutions.

7.6 Pending Evidence and Future Work

Evidence base status. As of 2026-04-10, the empirical evidence base for this paper is populated. Section 5.6 reports telemetry from the closed `stage2_derivation_seam_hardening` run against frozen sha256-snapshotted implementation files, and the deterministic verifier chain reproduces the closure-time verdict on re-run. Section 5.7 reports the live catch grammar self-audit with file-level artifacts already on disk. The two subsections report different classes of evidence for different claims and both are required for the paper’s argument. The Rule 1 Auditor Prompt spec’s mandatory discrimination test gate remains an explicit non-claim: it is not asserted by the paper as written, and it must not be wired into a live run until that gate passes. Post-publication research agenda items follow in the next paragraph.

Post-publication research agenda. The immediate external agenda is replication on other systems, stronger capital-efficiency measurement, and a cleaner principal abstraction layer. A more ambitious line is governance self-hosting under rule-bound independence: agents proposing changes to the governance contract while remaining subject to a deterministic check that the revised contract preserves the principal-independence invariant. The terminal-state direction is attestation rather than proof—external verification of compliance by a structurally separated layer, on the model of institutional audit rather than mathematical correctness. That framing does not lower the bar; it locates the bar in a place where empirical evidence can engage with it.

7.7 Toward an Independent Verification Institution for Recursive AI (Speculative)

The previous sections used the external-audit profession as a historical anchor (Sections 3.3 and 7.4), named a liability gap in the analogy (Section 7.5), and noted that the terminal-state direction is rule-bound independence rather than proof-theoretic closure (Section 7.6). Those moves are backward-looking and conservative by design. This section is the forward-looking complement: a compressed sketch of what an institutional architecture for recursive-AI verification could look like if the audit analogy is followed to its natural endpoint. Nothing in this section is supported by the empirical evidence in Section 5, and nothing in it is load-bearing for the T1–T4 claims. It is included as a direction for continuation and as a target for criticism, not as a design specification; the specific research questions that would need to be answered to instantiate any of it are named at the end of the section as future work.

The historical arc the audit analogy points at compresses to four moments: the Pujo Committee hearings (1912–1913) established that concentrated financial self-reporting was not measurable by the principals who needed to measure it; the Securities Exchange Act (1933–1934) interposed a structurally separated verification layer and codified a public rule set to enforce; the Big Four audit profession grew into that layer and supplied the triple of public rules, fail-closed verdicts, and principal-signed attestation; and Sarbanes-Oxley (2002), following Arthur Andersen’s collapse, added the two primitives the original framework had missed—a liability framework with real teeth and a structural firewall between audit and advisory revenue. The recursive-AI analog this section sketches takes the same shape, reduced to four named architectural elements plus one non-negotiable structural constraint.

A public, versioned rule library. The GAAP analog for recursive-AI verification would be a public, versioned catalog of falsification-gate specifications maintained outside any single laboratory and updated as new failure classes are discovered. The deterministic-gate contract described in Section 4.2 is the primitive shape this library would formalize. Without such a library, the

independence property cannot be audited by a third party, because each verification run is bespoke.

A principal-signed attestation instrument. The partner-signature analog is the structured, machine-verifiable attestation payload that names the rule library version applied, the failures found, the verdict, and a principal identity. The deterministic gate payload is already close to this shape in rudimentary form; the generalization is to make it cross-institutional.

A structurally independent verifier class. The Big Four analog is organizational rather than technical—institutions whose only product is attestation against the public rule library, whose credibility depends on their independence from the agents they verify, and whose compensation does not create a stake in the outcome. This is the element the paper cannot instantiate, but the slot must be named.

A non-negotiable audit/advisory firewall. The Arthur Andersen failure mode (2002) collapsed audit credibility because verification and advisory revenue were co-located in the same firms, and the independence property eroded under commercial pressure. The recursive-AI analog is explicit: the same institution cannot sell both verification against the public rule library and agent development or model training to the entity whose agents it verifies. This is not a design preference; it is the constraint the historical record shows is the single point of failure for an architecture of this shape. Any proposal that relaxes it is proposing the Andersen failure mode.

The liability / independence / rotation bundle—the unresolved institutional constraint. The audit profession’s robustness under adversarial pressure derives from a bundle of three primitives—statutory liability for false attestation, auditor-independence law (partner rotation, prohibited client relationships, peer review), and the compliance apparatus Sarbanes-Oxley layered on top. None of these has a settled analog for recursive-AI verification, and this section does not claim one. They are bundled here as a single unresolved institutional constraint because the paper’s current evidence and argument cannot distinguish between them; the constraint is named so that the gap is visible rather than smoothed over.

Continuation as future work. Three research questions follow directly from the sketch and would need to be answered before any part of this architecture could be instantiated. First, what primitives did the SEC and the early audit profession miss that only became visible in later crises—1929, the 1970s inflation-accounting debates, 2002 Enron—and do any of those missed primitives have obvious analogs in recursive-AI verification that the current architecture is vulnerable to? Second, what is the formal analog of auditor-independence law for agents, specifically when agents under verification and agents performing verification may share training data, architectural lineage, or evaluation history, and under what formal criterion should a verification run be re-audited by a second institution? Third, what form should the liability primitive take—statutory exposure, reputation capital, or internalized verification—for an institution class that does not yet exist, and is it bootstrappable from reputation before statute catches up? These are framed as research questions rather than design specifications because the answers will determine whether the institution class sketched above is viable and what its specific form should be.

8 Conclusion

The central claim of this paper is that agency cost—not raw capability—is the binding constraint on recursive AI systems in this implementation. When the same optimizing process generates output and evaluates that output, the system acquires an incentive to fabricate compliance rather than

produce real progress. The M-Form response is to separate generation from evaluation with a deterministic enforcement floor that the agent cannot negotiate away.

The contribution is not a promise of perfect alignment. It is a narrower and more defensible claim: hard-gated principal separation can bound the most destructive form of recursive self-deception, and it can do so in a way that is auditable and reproducible. The same specification gaming pattern—satisfying the letter of the evaluation contract while violating its load-bearing intent—was independently observed at the evaluator layer, the kernel layer, and the organizational layer of this system, suggesting that the adversarial gradient is a property of loop topology rather than of any particular substrate.

The architectural direction this paper points toward is not formal correctness but rule-bound independence—external attestation of compliance by a structurally separated verification layer, on the institutional model of independent audit; Section 7.7 sketches, speculatively, what an institution class in that shape could look like, and the paper’s claim is that the direction is the right one to continue, not that the specific form is known. The evidence is real but scoped. The architecture works in one system; it is not yet a universal law supported by broad replication. That scope limitation does not diminish the finding. If the same governance logic recurs across substrates and across layers because the underlying principal-agent problem recurs there too, then recursive AI improvement is not just a capabilities problem. It is also a firm-design problem. The M-Form is the current answer this system has for that problem.

Acknowledgements

The historical vocabulary that makes Sections 3.3, 7.4, and 7.7 possible—the Pujo Committee hearings, the post-1933 construction of the audit profession, the Arthur Andersen / Sarbanes-Oxley failure-and-response arc, and the framing of independent attestation as an institutional primitive rather than a technical one—was taught to me by Tom Nicholas at Harvard Business School. The mapping from that history to recursive-AI verification, and any errors in the mapping, are my own. I am grateful for a classroom that treated institutional architecture as a first-class object of study; this paper is an attempt to apply that lens to a governance problem that did not exist when the institutions in question were built.

Data and Code Availability

The replication package for this paper is bundled with the paper source at <https://github.com/sparckix/ztare/tree/main/papers/paper4>. It contains (i) the frozen run telemetry cited in Section 5.6—the closing status record, the event stream, and the deterministic verifier report for the reference build-pipeline program—together with reproduction instructions and the sha256 fingerprints of the implementation snapshot at closure; and (ii) the deterministic enforcement artifacts cited in Section 5.7—the non-LLM profile check and its regression pair, the pre-run guard wired into the experiment runner, the quote-locality sidecar and its regression pair, the catch grammar corpus, the Rule 1 Auditor Prompt specification with its mandatory discrimination gate, and the load-bearing probe record with the full dated demonstration chain. A provenance table mapping each epistemic label used in Sections 5.6–5.7 to its corresponding source file is included alongside the paper source. Papers 1, 2, and 3 in the four-paper arc are available on SSRN at the abstract IDs listed in the references; their replication artifacts are in sibling directories of the same repository. Readers who want to audit the numbers, verdicts, and artifact chain reported in

Sections 5.6 and 5.7 against primary sources can do so directly from the repository.

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